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Machine learning-based inference of oxidative stability of vegetable oils

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ABSTRACT

The induction period (IP, obtained via standard conditions and a commercial instrument) is nowadays one of the most common tools to investigate oxidative stability in lipid samples. Despite the advantages and widespread use of this methodology in industry, the development of algorithms to predict IP values has been impaired by the scarcity of data linking the composition of the sample with the corresponding IP value. Addressing this need and considering the persistent difficulties in bridging experimental and theoretical approaches related to edible fats, this paper reports on the development of a machine learning approach based on extreme gradient boosting (XGBoost) that was trained using a biodiesel database and then fine-tuned using a vegetable oils database, in both cases linking composition to their corresponding stability (via rancimat). The resulting model was able to predict accelerated oxidative stability of vegetable oils with a mean average error of only 1.37 h, one of the lowest ever reported. Beyond demonstrating the ability of our model to provide meaningful predictions, this report highlights the importance of incorporating oxidation-related chemical descriptors and the utility of using associated databases and transfer learning to support the development of accurate machine learning models.

1. Introduction

Oxidative stability is a fundamental chemical property of lipid-based formulations (Aparicio et al., 1999; Durand et al., 2025b; Musakhanian et al., 2022), as it serves as a proxy for the resistance of these samples to degradation due to exposure to oxygen (Bharti et al., 2024). Understandably, this process is particularly important to establish storage limits for food samples, as the oxidation of their lipids leads to significant changes in taste, flavor, texture, and ultimately safety (Ayres et al., 2023; Ayres et al., 2025; Wang et al., 2023). The oxidation of lipids, a radical-driven chain reaction, typically leads to the formation of undesirable low molecular-weight compounds (aldehydes, epoxides, or ketones) upon hydrogen abstraction from unsaturated fatty acids (FA) (Aghbashlo et al., 2021; Dike et al., 2025). As expected, the rate of lipid oxidation is determined by a combination of environmental factors (temperature, light exposure or oxygen) (Durand et al., 2025a), presence of antioxidants (Yoshida et al., 2023), presence of pro-oxidants (impurities or trace metals) (Choe and Min, 2006), and the composition of FA in the sample (Choe and Min, 2006; Wang et al., 2023; Yun and Surh, 2012). In general, these reports highlight the scientific gap between the vast number of methodologies available to experimentally determine oxidative stability and the limited capacity to make predictions related to the stability of lipid samples.

While several studies have attempted to model the chemical relationships between sample composition and oxidative stability, the developed frameworks typically rely on the use of model samples (Yun and Surh, 2012), specific experimental approaches (Tian and Dasgupta, 1999; Yoshida et al., 2023), the effect of antioxidants (Dike et al., 2025; Shahidi and Zhong, 2010b), the use of simplified unsaturation metrics (Bolonio et al., 2015a; Knothe and Dunn, 2003; Li et al., 2018; Pizzimenti et al., 2023; Redondo-Cuevas et al., 2018; Tsao et al., 2021), or the calculation of specific parameters such as the bond dissociation energy (BDE) (Murru et al., 2021). Considering the complexity and inherent variability of these systems (Li et al., 2013), it is important to highlight the recent use of various machine learning (ML) models in fields associated to oxidation reactions, including predicting flavor changes due to lipid oxidation (Chen et al., 2018), modeling lipid oxidation in emulsions (Liu et al., 2025), or monitoring lipid oxidative damage (Yi et al., 2025). Notably, and despite the progress made in biodiesel research (Aghbashlo et al., 2021; Barradas Filho et al., 2015; Bukkarapu and Krishnasamy, 2022; Díez-Valbuena et al., 2024; McCormick et al., 2007), there are no reports describing the inclusion of oxidation-related

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descriptors to predict metrics related to stability for edible fats. This issue can be attributed to the scarcity of literature reporting both sample composition and oxidative stability (Musakhanian et al., 2022), specially under accelerated conditions.

Addressing this need and considering the persistent difficulties in bridging experimental and theoretical approaches related to edible fats, this paper reports on the development of a series of ML approaches based on extreme gradient boosting (XGBoost), linking fatty acid composition of vegetable oils with their corresponding oxidative stability, determined using an established Rancimat method. To accomplish this goal, and taking advantage of strong generalization and transfer learning capacity of XGBoost, a model was first trained using an augmented version of a database released by Suvarna *et al.* (Suvarna et al., 2022), which provided a link between a chemically aligned source domain (fatty acid methyl esters) and their corresponding accelerated oxidative stability. The model was then fine-tuned, using transfer learning, using an *ad-hoc* database linking oxidative stability of vegetable oils with their corresponding fatty acid composition. While the model uses fatty acid percentage composition (from various vegetable oils) as its compositional input, it implicitly considers other components (including native antioxidant content) to make stability predictions. The current framework integrates several curated compositional datasets, experimentally measured oxidative stability values (Rancimat), chemically derived molecular descriptors, and robust machine learning algorithms. The results from this two-stage modeling approach not only shine light on the intricacies of complementary chemical descriptors that affect oxidative stability but also provide a path forward to enable predictions related to edible oils, a domain where the availability of data (linking composition and stability) is much more limited. More specifically, the presented approach and associated algorithm provide a simple way for researchers to estimate oxidative stability of oils, even if the selected composition not specifically included in the database.

2. Materials and methods

2.1. Reagents

All reagents used for the experiments were of analytical grade, stored under refrigeration, and used as received. All fatty acids used were purchased from Sigma-Aldrich (Saint Louis, MO) and included C14:0 (myristic acid), C16:0 (palmitic acid), C18:1 (oleic acid), C18:2 (linoleic acid), C18:3 (linolenic acid), C20:1 (gondoic acid), and C22:1 (erucic acid). These fatty acids were used to prepare synthetic mixtures, representative of naturally occurring oils (Díez-Valbuena et al., 2024) but without any of the natural components (phenolic contents, endogenous antioxidants, and pro-oxidant metals) that could influence their oxidative stability (Baiano and Del Nobile, 2016; Pokorný, 2007).

2.2. Rancimat analysis

The oxidative stability of fat samples was investigated via accelerated oxidation experiments using an 892 Professional Rancimat (Metrohm; Riverview, FL). This test is widely used in industry and was performed by heating an oil sample (3.0 g) at constant temperature (110 °C) and under a constant stream of air (20 L/h), bubbled in the sample. This stream of air is used to carry secondary oxidation products from the sample into a polystyrene vessel containing distilled water, thus gradually increasing its conductivity. The resulting oxidation stability was determined by measuring the induction period (IP), by manually identifying the inflection point of the second derivative of the conductivity *vs* time trace. It is important to state that while the specific experimental conditions selected (high temperature, air flow, etc.) may affect the oxidation mechanism, potentially leading to errors in the shelf-life predictions (Farhoosh, 2007), the method is simple, reproducible, and provides results that align well with other methods (Aktas et al., 2023; Bär et al., 2021; Bell et al., 2014; Pawar et al., 2014) that can complement the evaluation. Additional details of the instrument, the interpretation of the IP, and applications can be found elsewhere (Aparicio et al., 1999; Bär et al., 2021; Farhoosh and Hoseini-Yazdi, 2014b; Focke et al., 2016; Tinello et al., 2018).

2.3. Databases

The database utilized to train the primary model was initially compiled by Suvarna et al. (Suvarna et al., 2022) and contains 217 datapoints comprising biodiesels' FAME profile and the corresponding oxidative stability values (represented as IP values). This database aimed to enable predicting biodiesel compliance with ASTM (ASTM International, 2008) and EU (European Committee for Standardization, 2003) oxidative stability limits. Sample composition in the database includes the methyl esters of C14:0 (myristic acid), C16:0 (palmitic acid), C18:1 (oleic acid), C18:2 (linoleic acid), C18:3 (linolenic acid), C20:1 (gondoic acid), and C22:1 (erucic acid). To improve our models' robustness and predictive capabilities, we expanded the database by conducting a literature search to collate additional datapoints under similar experimental conditions (only studies providing both quantitative composition and experimental oxidation results for the same sample were considered). This approach improved the diversity of the dataset while expanding the number of samples to 371 unique datapoints, which was considered an adequate starting point. That said, a current limitation in the computational analysis of lipid oxidation is the scarcity of comprehensive datasets containing detailed composition profiles and experimentally measured IP in a relevant range. This is because most studies conduct compositional characterization or oxidative stability tests of lipid-based matrices, rather than both. It is important to note that the transesterification process (used to produce biodiesel from oils) does not change the profile of fatty acids, a consideration that, according to Barradas Filho (Barradas Filho et al., 2015), could allow the prediction of IP from the composition of the oil (before the transesterification). According to preliminary experiments, however, our model was unable to make meaningful predictions of the accelerated oxidative stability of vegetable oils using knowledge exclusively derived from biodiesel (Suvarna et al., 2022). This finding was attributed to the obvious differences in reactivity between the samples (biodiesel > vegetable oil / FAME > FA), even when their fatty acid profile was comparable. Moreover, this finding was somewhat expected, as even the small differences in reactivity between cis-linoleic acid (BDE = 328.0 kJ mol^{-1}) and trans-linoleic acid (BDE = 334.8 kJ mol^{-1}) have been reported to influence the rate of oxidation (Li et al., 2019), reinforcing the need to consider comprehensive structure-derived descriptors in predictive models of oxidative stability.

Thus, we also collated a second, *ad hoc* database (321 unique entries from literature) linking the fatty acid composition of vegetable oils with their corresponding oxidative stability (also described using IP values). Representative samples were extracted from published studies across diverse oil types (Ciemniewska-Żytkiewicz et al., 2014; Farhoosh and Hoseini-Yazdi, 2014a; Ghosh et al., 2019; Mateos et al., 2006). As later discussed in more detail, the data collected from the literature was skewed towards either high (FA database) or low (FAME database) IP values (*vide infra,* Fig. 2). To address this imbalance and improve the model's generalizability, we employed a Compositional Neighbor-Guided Augmentation approach, which enabled expanding the datasets to 466 FA entries and 500 FAME entries. To provide an unbiased set of samples to test the model, the FA database was randomly split into a training/validation dataset (446 entries) and a challenge dataset (20 entries, 15 entries from the literature + 5 augmented entries) that was never used for training purposes. The split ratio provides representation across the IP ranges on the challenge dataset, while preserving most of the dataset (96%) for training/validation purposes. As later described, these databases were then curated to remove outliers, leading to a final FAME dataset of 487 entries and a FA dataset of 434 entries (information related to the criteria used to remove the outliers is

provided in the Supplementary Information).

2.4. Molecular descriptors

To integrate structural features into our models, constituent FAME and FA were represented in the Simplified Molecular-Input Line-Entry System (SMILES) notation, obtained from PubChem. Next, each SMILES string was converted into molecular objects using RDKit functions, which enable calculating a broad range of chemical descriptors (Pullen and Saeed, 2012), including structural, physicochemical, topological, and electronic descriptors (Table SI 1). In addition to those descriptors, literature-established metrics linked to lipid oxidation (Bolonio et al., 2015b; Dwivedi and Sharma, 2014; Iyer, 2017; Knothe, 2002) were also incorporated into the dataset. APE and BAPE indices were calculated using Eq. (1) and Eq. (2), where A_x represents the percentage of the corresponding fatty acid in the sample.

$$\mathrm{APE} = 2 \times (A_{C18:1} + A_{C18:2} + A_{C18:3} + A_{C20:1} + A_{C22:1}) \quad (1)$$

$$\mathrm{BAPE} = (A_{C18:2} + 2 \times A_{C18:3}) \quad (2)$$

Other descriptors, such as dipolar moment or bond dissociation enthalpy (BDE) were obtained via DFT calculations, following a procedure recently described (Dike et al., 2025). Briefly, all geometries were optimized DFT at the TPSS-D3/def2-SVP level (Grimme et al., 2016; Tao et al., 2003; Weigend and Ahlrichs, 2005). Harmonic vibrational frequency analyses were subsequently performed at the same level to confirm the nature of the stationary points and to obtain the thermochemical corrections, (Chamberlin et al., 2008; Marenich et al., 2009; Ribeiro et al., 2010) which were used to calculate BDE. Specifically, these values were obtained by subtracting the energy of the deprotonated acid from the energy of the fully protonated one. DFT computations were performed using Gaussian 16 (Frisch et al., 2019) and provide a maximum chemical accuracy of ±1 kcal/mol (Feller and Peterson, 2007). A similar approach was performed to calculate the dipolar moment for FA and FAME, descriptors that were considered complementary to those obtained from RDKit. A summary of the results obtained is available in Table SI 2 (FA) and Table SI 3 (FAME). To establish a chemically meaningful consensus for each multi-component lipid system, a weighted descriptor aggregation (Zhang et al., 2024) approach was applied. For each descriptor d, a weighted average was calculated using Eq. (3), where $FA_i\%$ is the normalized fractional value of the i-th FA in the mixture, and d_i is its corresponding descriptor value. An identical approach was also applied to the biodiesel database, containing FAME.

$$\mathrm{Weighted_d} = \sum_{i=1}^{n} (FA_i\% \times d_i) \quad (3)$$

2.5. Machine learning

The computational work presented in this manuscript was performed in the Palmetto Cluster at Clemson University (palmetto.clemson.edu). An NVIDIA Tesla V100 was used as a graphical processing unit (GPU); the node was set to 8 cores, and the RAM was adjusted to 16 GB. Leveraging its efficient predictive features (Ayres et al., 2024; Dike et al., 2025) and its ability to handle imbalances in the training dataset (Jiang et al., 2025), the XGBoost algorithm (open-source framework) was selected for this work and programmed using Python 3.13.5, XGBoost: 3.0.4 scikit-learn: 1.7.1, NumPy: 1.26.4 Pandas: 2.3.3. This ensemble algorithm consists of multiple decision trees, where subsequent trees are built sequentially considering the results of their predecessors, thereby improving predictive capabilities and overall speed (Dong et al., 2022; Sibindi et al., 2023). Additional advantages of XGBoost over other ML models include its ability to learn from relatively small datasets (Karthikeyan and Priyakumar, 2021; Liang et al., 2021) and to perform transfer learning by continuing the training of an existing model on new data (Zhong et al., 2022). In our context, this feature allows the underlying chemical knowledge acquired in one domain (FAME) to be transferred to make predictions on a chemically related domain (FA), thereby addressing data scarcity. Fig. 1 provides a summary of the overall process applied to develop the primary FAME model, which was fine-tuned using a FA database into a FA model, ultimately capable of predicting the IP of vegetable oils.

It is important to note that a similar approach was recently applied to transfer knowledge from FA to triglycerides and make thermal conductivity predictions (Li et al., 2025), specifically noting that the similarities in physicochemical properties between the two domains would only require small corrections in the descriptors used. In this work, the mean absolute error (MAE, Eq. (4)) and root-mean-square deviation (RMSE, Eq. (5)) were selected to evaluate the machine-learning models. In both cases, P_i and A_i represent the predicted and actual values, respectively.

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n} |P_i - A_i|}{n} \quad (4)$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} (A_i - P_i)^2}{n}} \quad (5)$$

The SHapley Additive exPlanations library (SHAP, v0.48.0) was used to develop feature importance plots, which help understanding what features contribute the most to the intricate, complex relationship between sample composition and IP (Venegas-Reynoso et al., 2025; Winter, 2002).

3. Results and discussion

The following paragraphs describe the rationale used to develop a machine learning model to predict the oxidative stability (IP, via an accelerated method) of vegetable oils (FA model). The described approach specifically addresses the variability and inherent scarcity of literature data linking composition and IP, which has (up until now) precluded the development of reliable prediction models for oxidative stability of vegetable oils. Supporting this statement, and following best practices for limited datasets (Vabalas et al., 2019), a preliminary XGBoost model was trained using only the information contained in the FA dataset (n = 321), and a 5-fold cross-validation strategy. The predictive capacity of this preliminary model was then assessed using a challenge dataset (n = 20), never seen by the model. This approach, that follows the traditional workflow, only showed limited learning (cross validation $R^2 = 0.24$, MAE = 2.48, Figure SI 1A) and very limited predictions capabilities on the dataset never-seen by the model ($R^2 = 0.21$, MAE = 2.73, Figure SI 1B). Considering these limitations, this report describes a broader approach, that relies on the development of a primary model (FAME model) that was pre-trained using accelerated stability data of biodiesels and then fine-tuned (via transfer learning) to account for the corresponding differences in reactivity and composition of vegetable oils. The variability of the oxidative stability data collected in both datasets is initially presented, to highlight both the complexity of the problem and the criteria applied to implement the transfer learning process.

3.1. Overall oxidative stability trends

To determine the predictive value of a simple analysis of the FA database, the oxidative stability (expressed as IP, hours) was initially investigated as a function of the corresponding composition (FA, %). As it can be observed in Fig. 2A, these samples show a large dispersion across a wide IP range (3.5 – 17.5 h), where only general trends in oxidative stability are visible. In terms of the composition of FA reported, these samples showed an average of 42 ± 20% C18:1 (oleic acid), 34 ± 20% C18:2 (linoleic acid), 12 ± 8% C16:0, 3 ± 7% C18:3,

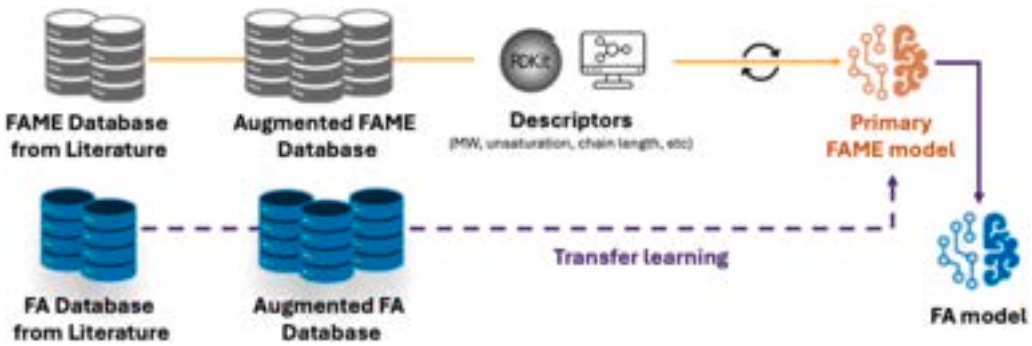

Fig. 1. Schematic diagram describing the process implemented to train the primary model (based on FAME). A FA database and transfer learning was subsequently used to develop the FA model, to predict oxidative stability (IP) of vegetable oils. For the sake of simplicity, only the procedure utilized to train the FAME model is specifically described.

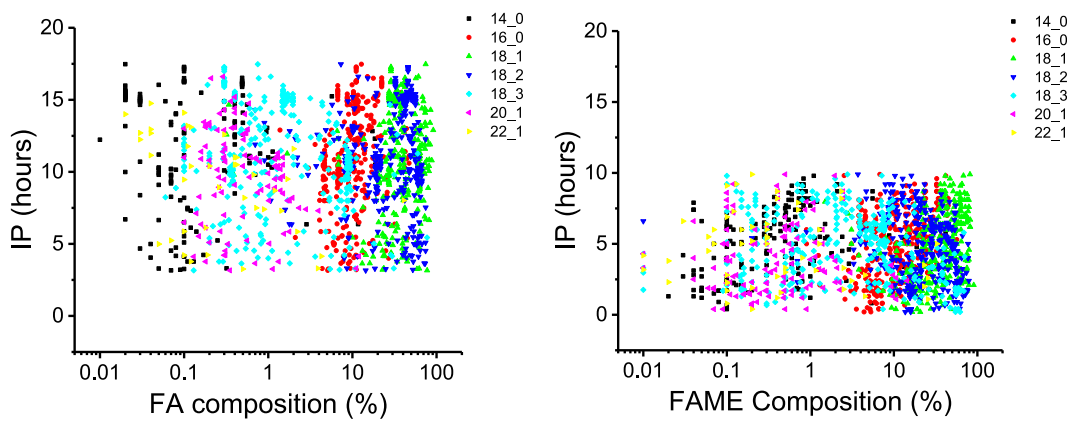

Fig. 2. A: Scatter plot overview of the FA database collected from literature (n = 321), showing the dependence of the oxidative stability (IP) and the FA composition. Fig. 2B: Scatter plot overview of the FAME database collected from the literature (n = 371), showing the dependence of the oxidative stability (IP) and the FAME composition.

and less than 1% of C14:0, C20:0, and C:22:1. Although this relative abundance may suggest a primary role of oleic and linoleic acids in the initiation of the lipid peroxidation reaction (Pullen and Saeed, 2012), it is important to note that these are only general trends and do not represent a comprehensive view of the entire dataset. For instance, while many samples containing > 15% of C14:0 showed high stability (IP = 12 ± 1 h) and many samples containing significant amounts of PUFA (>70% of C18:2 and C18:3) featured relatively low stability (IP = 6 ± 3 h), many of the most stable samples (IP ≥ 17 h) contain at least 75% of C18:1 and C18:2. This apparent contradiction between chemical reactivity and sample stability can be attributed to the complexity of intermolecular interactions and the variable presence of phenolic compounds (such as tocopherol (Shahidi and de Camargo, 2016)) typically found in these samples. As an additional possibility to identify underlying patterns, the dataset was divided into three different IP bins of low (<4 h), mid (4–7 h), and high (≥7 h) stability. The results (provided as Figure SI 2A) only show marginal differences in terms of the effect of lipid composition, where most samples feature a significant percentage of unsaturated fatty acids (C18:1 and C18:2) (Gutiérrez-Luna et al., 2022; Redondo-Cuevas et al., 2018; Shahidi and Zhong, 2010a; Yun and Surh, 2012) and where the increases in IP can only be associated with slight differences in C18:3 (α-linolenic acid), a particularly susceptible FA (Mavlanov et al., 2025).

Also to determine the potential predictive value of the FAME database (Suvarna et al., 2022), the oxidative stability of biodiesel was investigated as a function of the lipid composition (FAME, %). As it can be observed in Fig. 2B, the results from these samples also show a large dispersion across the investigated IP range, where only general trends are visible. For example, a large percentage of saturated FAME (such as methyl myristate or methyl palmitate) can be generally associated with longer induction periods, in agreement with the absence of a susceptible carbon-carbon double bond, which decreases the chance for hydrogen abstraction and leads to improved oxidative resistance (Bielecka et al., 2023; Yao et al., 2024). The scatter plot further reflects the broad oxidative response relative to the abundance of methyl oleate, formed by transesterification of oleic acid (C18:1), a significant constituent of many natural fats and oils (as shown in Fig. 2A). As evidenced by the DFT calculations later discussed, this finding was somewhat expected, as C18:1 features an intermediate reactivity that results in a wide range of IP values. Other MUFA identified in this work (C20:1 and C22:1) also exhibit similar intermediate oxidative behavior in these samples, albeit their contribution is less evident due to their lower relative abundance. It is also remarkable to observe that despite the poor oxidative stability of PUFAs (McCormick et al., 2007; Wang et al., 2023), samples containing significant amounts of C18:2 and C18:3 can still feature moderate stability. Following the analysis performed using the FA dataset, the FAME dataset was also divided into three different IP bins. The results (Figure SI 2B) show the progressive increase in IP as the median percentage of C16:0 (methyl palmitate) grows from 7% (in the <4 h bin) to 16% (in the ≥7 h bin). In contrast, samples featuring progressive increases in methyl esters of unsaturated fatty acids (C18:1 and C18:2) showed a progressive decline in IP, which is consistent with the established notion that increased unsaturation decreases oxidative stability. Despite these general findings, these complex and non-linear patterns illustrate the difficulty in defining a generalized concept across the entire datasets, precluding their analysis via conventional means, and highlighting the need to implement more advanced predictive tools, such as ML.

3.2. Data augmentation

It is well known that not only the size but also the distribution of the database used for training purposes could have a significant effect on the accuracy of the final predictions (Al-Jubouri et al., 2025; Liang et al., 2021). Unfortunately, low IP samples dominated the initial FAME database (<4 h, n = 159), while mid IP (4–7 h) and high IP (≥7 h) samples were comparatively underrepresented, as shown in Fig. 2B. This imbalance, which can be traced back to the objective of ensuring compliance of these samples with biodiesel regulations, was particularly important, as it affected the most chemically relevant data for predictions in vegetable oils. To address this issue and improve the generalization of our model, we preliminarily evaluated the effectiveness of several augmentation approaches (including SMOGN (Branco et al., 2017; Yin et al., 2025)) and ultimately decided to implement a Compositional Neighbor-Guided Augmentation approach (Islam et al., 2022), to introduce controlled variability in percentage compositions. Here, synthetic samples were generated within our established IP bands by interpolating between nearest-neighbor compositions and applying Dirichlet-based perturbations (Pal and Heumann, 2022; Zhao et al., 2025) on the composition percentages while introducing Gaussian noise into the target IP. Using a multistep process based on the parameters described in Table SI 5, generated compositions were clipped within empirical quantile ranges and duplicates were then removed. This process led to the addition of 145 synthetic FA datapoints and 129 FAME synthetic datapoints to the corresponding databases, leading to a distribution (Fig. 3A and B) that was considered more adequate for our algorithm to learn composition patterns related to oxidative stability and ultimately understand the chemical differences between FAME and FA.

3.3. Primary FAME model

The augmented FAME dataset (n = 500, Fig. 3B) was used to develop the framework required to compute the relationship between FAME composition and oxidative stability of biodiesel samples. Toward that goal, SMILES notations for each FAME component were added to the dataset, along with a series of descriptors, computed considering their corresponding compositional percentages. Next, a hyperparameter search based on cross validation-based was conducted to obtain optimal combinations (ranges explored highlighted in Table SI 4). Following best practices for limited datasets (Vabalas et al., 2019), the primary FAME model (Fig. 1) was then trained using the augmented FAME database and a 5-fold cross-validation strategy. This approach involved randomly splitting the augmented FAME dataset (n = 500) into five groups and then sequentially using four of those groups for training and the remaining group for validation purposes. For each of the folds, the training process was allowed to run until convergence was observed. A preliminary analysis of the resulting FAME model showed MAE values of 1.13 h and RMSE of 1.49 h and while these values are only slightly better than those obtained with the model trained using the literature database (n = 371, MAE=1.28 h, RMSE=2.01 h), the augmentation strategy was particularly effective for the predictions for the high stability band (> 7 h), where a significant reduction in the MAE was obtained (from ~1.74 h to 0.94 h). Additional information related to the impact of the data augmentation strategy on each section of the IP range is available in Table SI 6. More importantly, these results allowed the identification and subsequent exclusion of 13 outlier samples (2.6% of the dataset) and addressing inconsistencies between the IP relative to their FAME profiles. These inconsistencies were attributed to potential errors reported in the original papers, issues generated during the data retrieval, or the inclusion of samples with unusual composition. Thus, a final FAME model was then re-trained using the curated FAME database (n = 487 entries), resulting in an additional improvement in its predictive performance (R^2=0.69, MAE=1.02 h, RMSE=1.31 h, Fig. 4A). While it is important to acknowledge that an average error of ~1 h remains relatively high when compared to the reproducibility of typical results based on the Rancimat method, these deviations can be reasonably attributed to the size of the dataset, unaccounted differences in matrix composition, and variations in the collated interlaboratory literature data. Despite these limitations, the herein described model, over the previously described approaches is the explicit inclusion of oxidation-related molecular descriptors (Table SI 1) allowing more accurate predictions of stability based on IP values and potentially extending the applicability of the model to other problems, such as determining the appropriate antioxidant dosage for these samples (Knothe, 2007; Kumar and Sharma, 2015). Moreover, the described strategy not only provides good estimations of oxidative stability based on composition but also represents a significant advance in terms of the development of machine learning methodologies for biodiesel applications. For instance, Barradas Filho et al. (Barradas Filho et al., 2015) developed artificial neural networks and regressor models to predict induction periods from fatty acid composition in biodiesel, reporting an RMSE of ~0.9 h. While slightly more accurate than our model, the size of the utilized dataset (only 98 samples) may be insufficient to broadly capture the natural variability of biodiesels and avoid overfitting. In another study, Yoshida et al. (Yoshida et al., 2023) employed a classical linear regression method incorporating chemiluminescence data, alongside fatty acid composition and tocopherols, to predict the IP of 13 edible oils. Although effective (adjusted R^2 values >0.9), this approach was also constrained by the limited number of analyzed edible oils and relied on specialized chemiluminescence measurements.

An additional advantage of the selected XGBoost model is that it can provide interpretability, enhancing our understanding of the

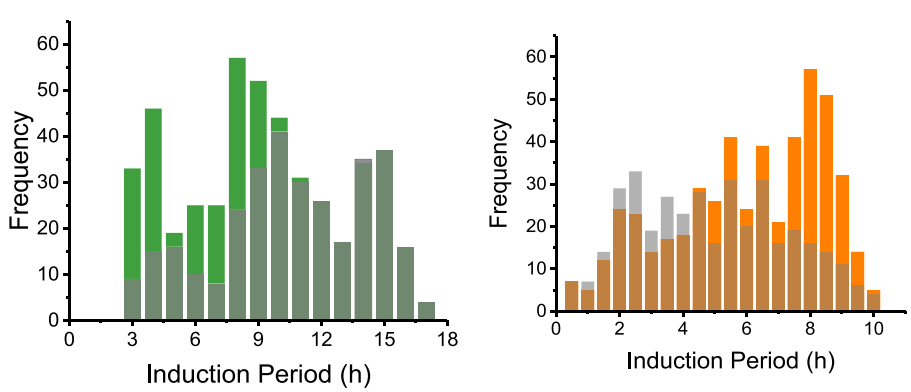

Fig. 3. A: Histograms showing the sample distribution in the FA database containing only literature-reported entries (■, n = 321) or entries after implementing the augmentation strategy (■, n = 466). Fig. 3B: Histograms showing the sample distribution in the FAME database containing only literature-reported entries (■, n = 371) or entries after implementing the augmentation strategy (■, n = 500).

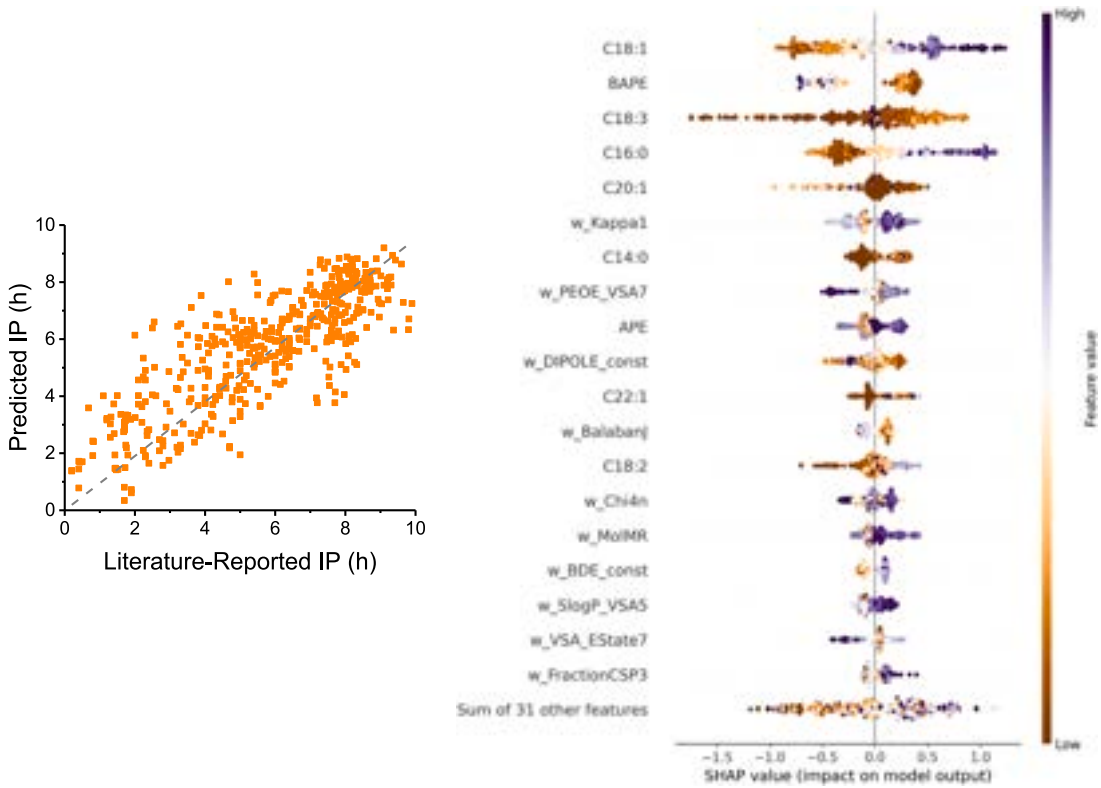

Fig. 4. A: Relationship between the model-predicted and the literature-reported IP values for the five validation groups (curated FAME database, n = 487), after convergence of the training process. Fig. 4B: Beeswarm plot of the SHAP values for deployed features and their impact on the primary FAME model.

relationship between FAME composition and oxidative stability of biodiesel samples. Hence, a SHAP analysis (Ponce-Bobadilla et al., 2024; Rodríguez-Pérez and Bajorath, 2021) was performed, considering that such an analysis is bounded by the model architecture and descriptor representation (composition-weighted molecular descriptors) (Rodríguez-Pérez and Bajorath, 2020). Briefly, chemical descriptors indicating high positive SHAP values (on the x-axis) can be related to improvements in oxidative stability (increased predicted IP). Conversely, descriptors featuring high negative SHAP values are linked to reduced oxidative stability (lower predicted IP). Accordingly, the SHAP values describe how our selected features influence the prediction of induction periods within our specified context.

It is important to note that the interpretability of the developed model is constrained by the selected feature representation (fatty acid composition and weighted molecular descriptors), and the chemical space represented within the curated and augmented databases. Therefore, the interpretation of the SHAP values presented in this study is directly related to its importance within the present modeling architecture and dataset, rather than implying universal chemical dominance in all lipid oxidation systems. As it can be observed in Fig. 4B, the Beeswarm plot of the SHAP values for the top 20 features corroborates the established relative contributions of fatty acids, which broadly align with established lipid oxidation chemistry (Pullen and Saeed, 2012). The results identify the % of oleate (C18:1) as the most influential factor in the oxidation of FAME systems, exerting an overall positive influence on stability (Duarte et al., 2018). Similarly, saturated ester palmitate (C16:0) also revealed a positive correlation, highlighting its stabilizing effect in biodiesels (Verghese and Saeed, 2025). In contrast, high percentages of polyunsaturated esters, particularly linolenate (C18:3), were associated with negative SHAP values yielding to shorter induction periods, which is consistent with lower bond dissociation energies and increased susceptibility to hydrogen abstraction (Table SI 3) (Verghese and Saeed, 2025). APE and BAPE values, which represent allylic and bis-allylic sites, respectively, were also identified as important oxidative predictors across our models to adversely affect the IP. Specifically, the highly reactive rate of bis-allylic CH_2 positions (featuring lower BDE (Murru et al., 2021)) is reflected in the SHAP analysis, where high BAPE values consistently correspond to lower predicted induction periods across the model. Topological indices such as Kappa1, Balaban J, and Chi4n reflect molecular acyclicity and chain linearity relative to branching (Hall and Kier, 1991; Saviñon-Flores et al., 2025), which affect the exposure of reactive sites. Previous oxidative studies (Sargis and Subbaiah, 2003) have highlighted that cis/trans configuration alters hydrogen accessibility, where cis fatty acids oxidize more readily than their trans counterparts. Also in line with previous studies (V et al., 2017), the BDE also demonstrated a positive contribution to IP, an effect attributed to its correlation with the corresponding activation energy for the initiation of the radical chain reaction.

3.4. FA model

To effectively predict the oxidation stability of vegetable oils, we adopted a two-stage transfer learning strategy that combined feature and model transfer, similarly to the approach employed by Li et al. (Li et al., 2025). In our case, the primary FAME model was fine-tuned with an *ad hoc* database linking FA composition for vegetable oils with their corresponding induction times. By computing the same dimensional molecular descriptors utilized in the FAME model (including oxidation-related descriptors), this approach ensured consistency and preserved chemical domain knowledge relevant to lipid oxidation. It is critical to note that a preliminary assessment of the capacity of the primary FAME model to predict induction periods of vegetable oils (prior to transfer learning) revealed its inability to make meaningful predictions (MAE = 5.44 h and RMSE = 6.59 h), as shown in Figure SI 4. This issue can be attributed to multiple factors, including the domain shift between biodiesel and edible oil systems (Yaşar, 2020), the heterogeneity of the matrix, the triglyceride forms of fatty acids in edible oils (Yin et al., 2011), and the presence of endogenous antioxidants (Yin

et al., 2011) that can skew the prediction. These results emphasize the importance of domain adaptation via fine-tuning, considering that the transfer of chemical knowledge is based on the chemical overlap between fatty acids and the corresponding fatty acid methyl esters (Knothe, 2005; Ramos et al., 2009). Therefore, we retained the pre-trained FAME model's architecture and leveraged continued boosting to extend its trees, enabling its adaptation to vegetable oils. Again, mimicking the process implemented to train the primary FAME model, a 5-fold cross-validation strategy (randomly shuffled with a fixed seed) was implemented to fine-tune this model, using 446 entries from the augmented FA database, until convergence (Fig. 5A). During this process, model performance was evaluated using out-of-fold (OOF) predictions, ensuring that each validation sample was predicted by a model that had not been trained on it, thereby preventing data leakage. The resulting FA model featured adequate prediction capabilities, as determined by the validation set over the 5 folds (MAE = 1.62 h and RMSE = 2.23 h, Fig. 5B).

To challenge the predictive capacity of the model and perform an unbiased analysis of the effectiveness of the transfer learning process, a set of 20 entries, randomly selected from the augmented FA dataset and never shown to the model, was processed by the FA model. As discussed earlier, 15 samples correspond to literature-reported samples (e.g., cold-pressed unrefined rapeseed oil (Roszkowska et al., 2015), canola oil (Castelo-Branco et al., 2016), poppy seed oil (Dedebas, 2024a), vanaspati samples (Farmani and Gholitabar, 2015; Tavakoli et al., 2019a), soybean oil and sesame oil (Hassanein and Abdel-Razek, 2012), and olive oil (Heidarpour and Farhoosh, 2018)). The challenge set was complemented by including five samples generated through the compositional augmentation strategy.

As can be observed in Fig. 6, a good agreement ($R^2 = 0.97$) and a low error (MAE = 1.37 h) were obtained between the predicted and literature-reported IP values. While these results indicate promising external predictive performance, we acknowledge the limited size of the challenge dataset, primarily due to the limited availability of data linking sample composition with the corresponding oxidative stability. To the best of our knowledge, these results represent a suitable demonstration of the ability of the proposed machine learning model to predict oxidative stability of vegetable oils using an accelerated test (Rancimat), spanning from more labile oils (such as poppy seed oil (Dedebas, 2024b)) to more stable oils (such as cooking oil (Tavakoli et al., 2019b)). Moreover, the error associated with our predictions is within the relative standard deviations for the instrument (1–5% or 0.2–0.6 h), for typical samples of vegetable oils (0.5 – 2 h for linseed oil, 7–12 h for palm oil, or 12–17 h for canola oil).

In order to demonstrate the advantages of the herein described approach, the same challenge dataset (n = 20) was presented to the latest version of ChatGPT (based on the GPT-5.3 architecture) and asked to estimate IP values for those vegetable oils. As can be also observed in Fig. 6, this conventional language model was only able to estimate

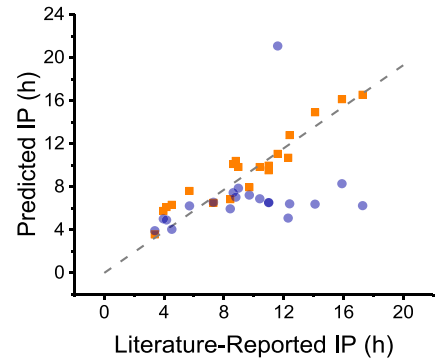

Fig. 6. Relationship between the predicted and the literature-reported IP values, for the challenge dataset (n = 20) of vegetable oils. Figure includes results from the FA model (■), trained using the curated FA dataset (n = 434) and results from ChatGPT (■), based on the GPT-5.3 architecture.

reasonable IP values (110 °C) for a few samples with IP < 9 h. While noting that these are rough estimates and should be treated as ±20–30% depending on antioxidants, refining history, metals, and the exact Rancimat temperature it also mentioned that the usual oxidative stability calculation weights C18:2 and C18:3 much more heavily than C18:1, and that Rancimat's IP is a relative stability measure rather than a fixed material constant, where higher linolenic content pushes stability down disproportionately.

3.5. Additional considerations

At this point, is important to bring up a few additional aspects that pertain to the overall project herein described. First, and in order to evaluate the impact of each of the steps described in the project (augmentation and transfer learning strategy), the performance of pertinent models was compared using metrics from either the training step or the predictions made by each model on the challenge dataset of vegetable oils. As it can be observed Table 1, none of the models trained using a single database (raw or augmented) was able to capture meaningful differences in oxidative stability. This can be attributed to the limited amount of information provided during the training step, regardless of the descriptors used. That said, the inclusion of molecular descriptors provided significant advantages on the predictive abilities of the models, aspect that can be attributed to the inclusion of chemically relevant information. As expected, the worst model was the one trained using the FAME database and used (without any fine tuning) to make predictions of vegetable oils, which despite the obvious structural similarities between FAME and FA, they feature clear differences in reactivity. Out of all the possible combinations, the best performance was obtained when all the descriptors (% composition and molecular

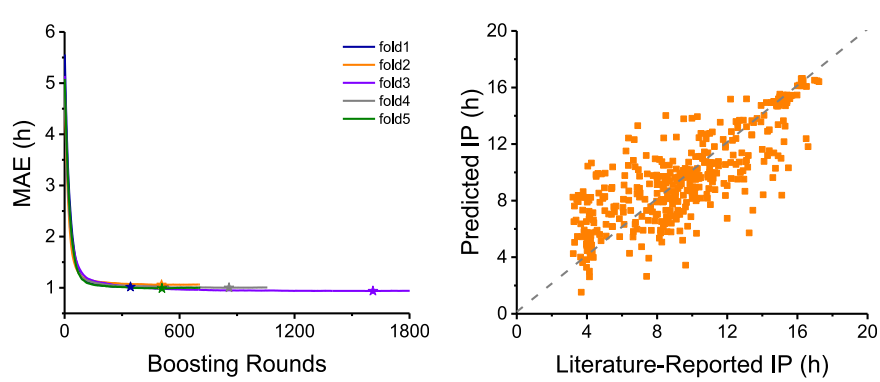

Fig. 5. A: Evolution of validation MAE during the training step using the augmented FA dataset (n = 446). The figure also shows the point of convergence (★) for each fold. Fig. 5B: Relationship between the model-predicted and the literature-reported IP values for the augmented FA dataset (n = 446).

Table 1
Comparison of the performance for models trained and/or evaluated on different datasets.

Training Dataset		Descriptors	Cross validation			Challenge dataset		
FAME	FA		R^2	MAE	RMSE	R^2	MAE	RMSE
-	Literature	Composition	0.24	2.48	3.11	0.21	2.73	3.48
-	Augmented	Composition	0.34	2.54	3.08	0.54	2.24	2.66
-	Augmented	Composition + Molecular	0.60	1.72	2.41	0.82	1.33	1.64
Augmented	-	Composition + Molecular	0.60	1.13	1.49	0.04	4.50	5.58
Augmented	Augmented	Composition + Molecular	0.65	1.62	2.23	0.97	1.37	1.54

descriptors) were used to train the primary FAME model and then fine-tune it using the FA dataset. These results highlight the advantages of transfer learning for this application, leveraging knowledge from a chemically related domain (FAME database, n = 487).

The second aspect that is worth mentioning is that a benchmark analysis of the selected XGBoost model against that of partial least squares (PLS) and Random Forest models was performed using identical 5-fold cross-validation splits and feature sets. As shown in Table SI 7, the PLS model displayed the worst performance (MAE =1.39, and R^2= 0.43), results that were attributed to the nonlinear relationship between FAME composition and oxidative stability. Although the random forest displayed considerable improvements (MAE =1.08, and R^2= 0.63), the XGBoost achieved the best overall performance.

Third, the results highlight the importance of having sufficient information to support the training process. As it can be observed in Fig. 7, the larger the number of examples used for training, the better the performance. That said, the performance of the described FA model (developed from the primary FAME model + transfer learning) was superior than that of a FA model trained from scratch using only the FA dataset. As a point of reference, and even when the entire FA database was used, the FA model trained from scratch only achieved a MAE value of 2.68 h, underscoring the advantages of combining multiple datasets. Although the performance for the transfer-learned model apparently plateaus after 130 data points when evaluated by MAE, it's important to emphasize that incorporating additional data improves the model's ability generalize to unseen cases and expands the coverage of IP prediction. Relying on only a small fraction of the points risks overfitting to the specific distribution of the limited training inputs, which may not be reflected by MAE alone. To verify this hypothesis, we compared the performance of the model trained with 30% of the data against the full dataset by examining the distribution of predicted IP values. As expected, the model trained with the reduced dataset produced IPs clustered within a narrow IP band, while the full model generated a broader and more realistic distribution that better captures the underlying variability of IPs across different samples of vegetable oils (Figure SI 5).

It is also important to consider that although the transesterification process does not alter the fatty acid profile of the vegetable oil samples

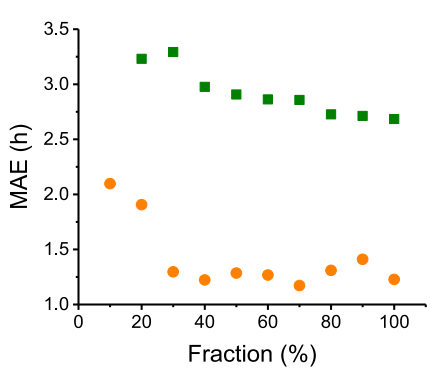

Fig. 7. Dependence of the MAE as a function of the fraction of the dataset used to either train a FA model from scratch (■, no transfer learning) or to fine-tune a primary FAME model via transfer learning (●). In this case, the performance of each model was evaluated against the challenge dataset (n = 20).

(Barradas Filho et al., 2015), it has been widely reported that the reaction leads to significant changes in the sample, including decreases in the concentration/activity of the natural antioxidants present in the sample (Medina-Juárez et al., 2000; Yildiz et al., 2025). Considering that such antioxidants would be expected to have a significant effect on the oxidative stability of the oils, mixtures of lab-grade FA were generated, in proportions that mimic common vegetable oils. These samples were analyzed using the Rancimat method and the resulting IP values compared to those predicted by the FA model. As can be observed in Table 2, even our best model significantly overestimated the stability of these mixtures, leading to predicted IP values that were (in average) almost 50 times larger than those experimentally obtained with the mixtures of lab-grade fatty acids.

These results are in line with a recent report stating that the presence of natural antioxidants play a determining factor in the stability of vegetable oils and has an effect that could even be more pronounced than the concentration of fatty acids (Farhoosh and Hoseini-Yazdi, 2014b). In our case, and although the total antioxidant capacity of the samples is not specifically computed, the contribution of both antioxidants/prooxidants present in the samples is implicitly included in the predictions, as the algorithm was trained linking vegetable oil compositions with the corresponding IP values. Moreover, these results not only highlight the importance of understanding the role of antioxidants in the stability of oils (Ayres et al., 2025) but also underscore the need to develop databases that provide a more comprehensive view of the (anti) oxidative chemistry of these samples.

4. Conclusions

This study described a machine learning approach based on extreme gradient boosting (XGBoost) to predict IP values (obtained via a Rancimat method) for biodiesel samples with an average error of 1.02 h. This primary model was trained using a FAME database, providing the ground knowledge related to lipid oxidation chemistry. The model was then fine-tuned via transfer learning using an *ad hoc* database linking fatty acid composition of vegetable oils with their corresponding oxidative stability (also obtained via Rancimat). In both cases, a data augmentation strategy was applied to increase the representativeness of the training set, specifically addressing the inherent scarcity of literature data linking composition and IP, and ultimately resulting in predictions of stability of vegetable oils with a MAE of only 1.37 h (on a challenge dataset). Beyond demonstrating the ability of the FA model to provide meaningful predictions related to edible oils, a domain where the availability of data (linking composition and stability) is very limited, our results set the basis to potentially extend the gained knowledge to other applications (Tachie et al., 2023; Tao et al., 2024; Wang et al., 2024). According to our results and despite having the capacity to search existing literature at a speed that humans will never match, commonly used machine learning tools (such as ChatGPT) do not yet have the capacity to make accurate predictions. On the contrary, our results not only provide a simple way to estimate IP values based on composition and quickly identify experimental outliers but also evidence the need to collect specific data related to the type/concentration of antioxidants in these samples, which would enhance the accuracy of future models.

Table 2
Comparison of the predicted vs experimentally obtained IP values for mixtures of lab-grade fatty acids.

Mixtures selected							Matching vegetable oil	Predicted IP (h)	IP Rancimat (h)
C14:0	C16:0	C18:1	C18:2	C18:3	C20:1	C22:1			
0.8	25	19	53	0	0	0	cottonseed	10.60	0.2
0.1	7.2	19	69	0	0.3	0	grapeseed	4.53	0.08
0	12	75	9	0.3	0	0	olive	10.75	0.2
1.6	45	40	9	0.2	0	0	palm	11.42	0.17
0	11	24	49	1.2	0.3	0.1	soybean	7.04	0.15
0	5	20	17	55	0	0	linseed	4.66	0.18
0	7	25	64	0.8	0	0	sunflower	7.41	0.18
0	9	43	43	0	0	0	sesame	7.89	0.18

CRediT authorship contribution statement

Garcia Carlos: Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Whitehead Daniel:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Justin Furgala:** Writing – review & editing, Data curation. **Jorge Barroso:** Writing – original draft, Software, Methodology, Conceptualization. **Dike Emmanuel:** Writing – review & editing, Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Ayres Lucas:** Writing – review & editing, Software, Data curation, Conceptualization.

Declaration of Competing Interest

Authors declare no conflicts of interest.

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Supporting information

This article includes information related to Outlier removal, Chemical Descriptors, Considerations for the FA preliminary model, Overall stability trends – cluster analysis, Hyperparameter search, Data Augmentation, Predictions using the primary FAME model, and the distribution of the IP predicted for the challenge dataset using a model trained from scratch using either 30% or the full FA database.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jfca.2026.109135.

Data availability

The data that has been used is confidential.

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